Resting-State fMRI-based Autism Spectrum Disorder classification

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01

Introduction

Definition

– Autism spectrum disorder (ASD) is a neurodevelopmental disorder characterized by deficits in social communication and the presence of restricted interests and repetitive behaviors [1].

– According to ICD-11 (WHO) [2]:

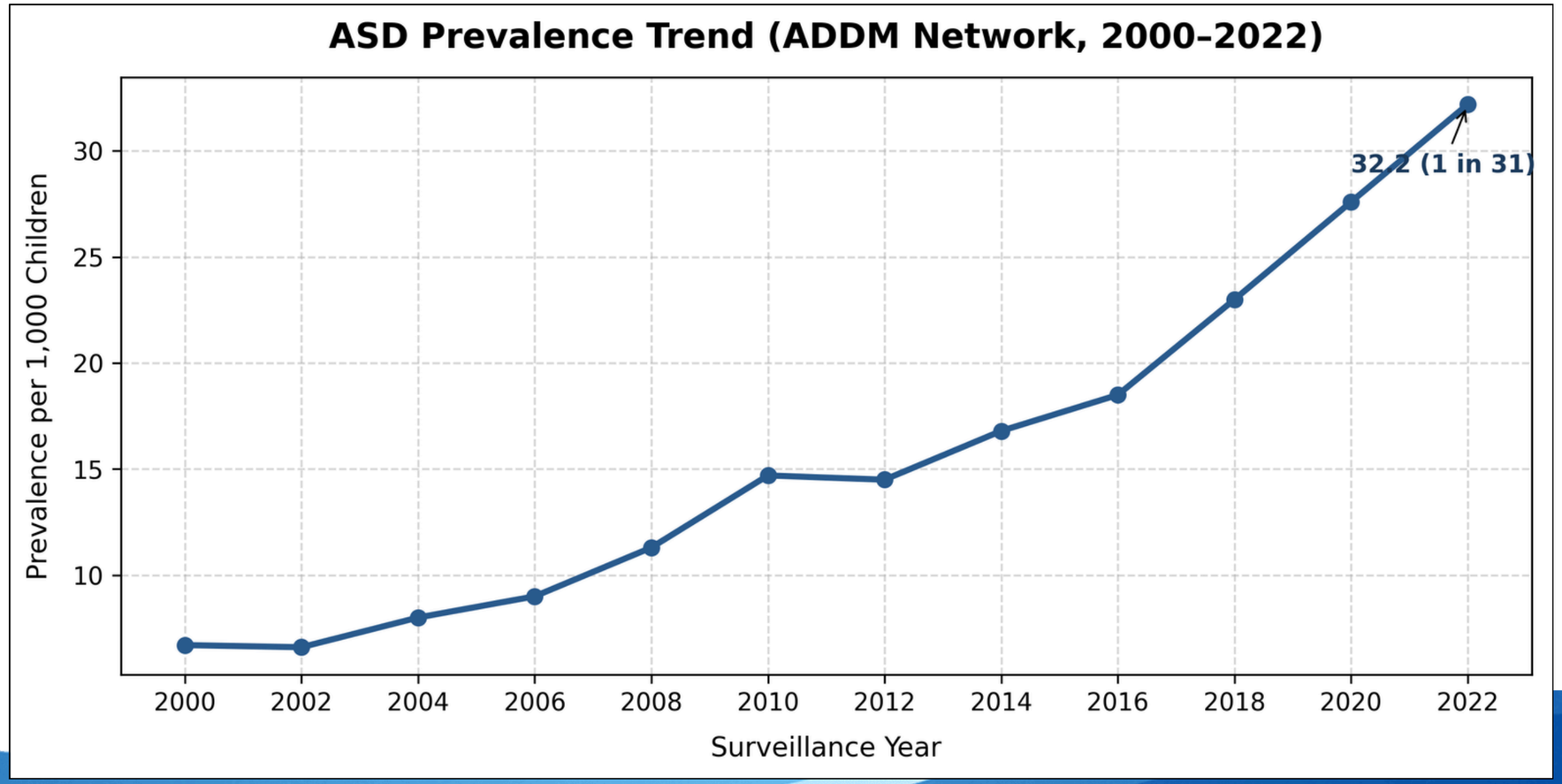
> “Autism spectrum disorder is characterised by persistent deficits in the ability to initiate and to sustain reciprocal social interaction and social communication, and by a range of restricted, repetitive, and inflexible patterns of behaviour, interests or activities.”

– A similar definition is established by the American Psychiatric Association in the DSM-5. [3]

Current Status

– Once considered rare, ASD is now estimated to affect approximately 1% of children. [4]

– In 2022, approximately 1 in 31 8-year-old children (3.2%) in the United States were identified with ASD. [5]



Current Diagnostic Limitations

– There is no proven biological meaurement (e.g., blood test or radiological scan) which can identity pathophysiological processes that can aid in the diagnosis or treatment of ASD at this time. [6]

– Diagnosis primarily relies on: [7]

- Developmental history
- Caregiver reports
- Direct behavioral observation
- DSM-5 or ICD-11 criteria

– ADOS-2 and ADI-R support clinical assessment but are not biological tests

Research question

– RQ1: Can functional connectivity distinguish individuals with ASD from typically developing individuals?
– RQ2: Do deep learning models outperform classical machine learning models?
– RQ3: Can graph neural networks better exploit the graph structure of brain networks?
– RQ4: Can XAI reveal brain regions and connections that consistently contribute to ASD classification?
– RQ5: Do the learned patterns generalize across imaging sites?

02

Dataset

ABIDE Dataset

– Large-scale, multi-site neuroimaging dataset for ASD research.

– Includes rs-fMRI, sMRI and phenotypic data; Publicly released ABIDE I and ABIDE II datasets: [8]

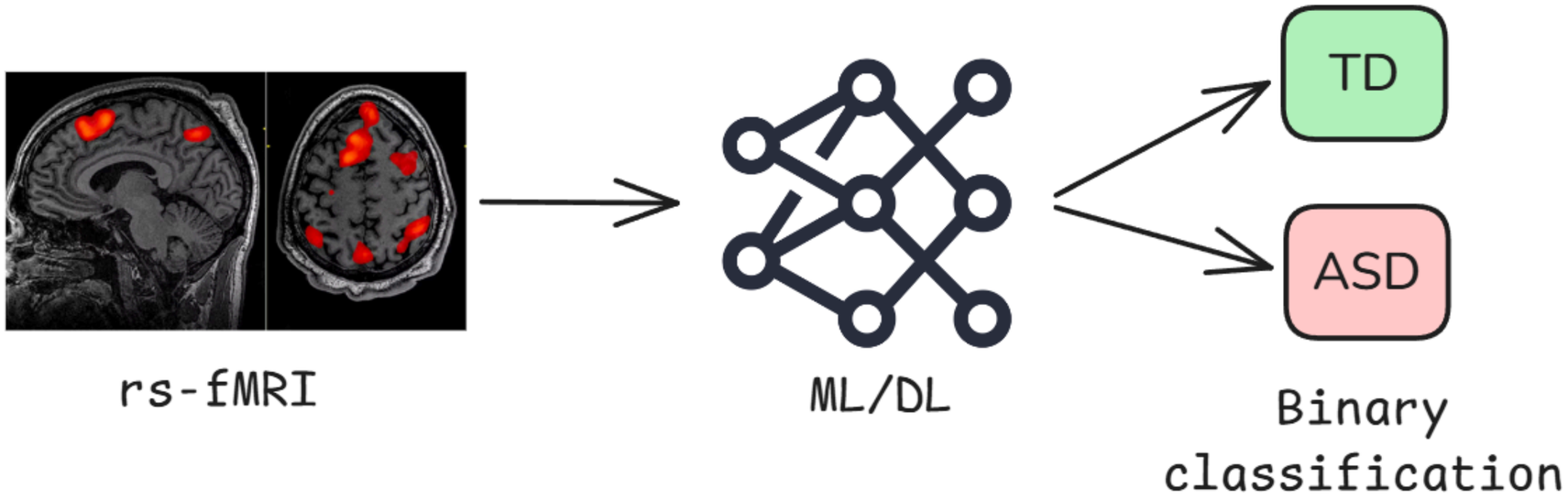
	ABIDE I	**ABIDE II**
Release	2012	2016
Sites	17	19
ASD / TD	539 / 573	521 / 593
Age	7–64	5–64

> **This study uses ABIDE I**

03

Methodology

Overall Pipeline



– Input: rs-fMRI (Resting state fMRI):
  - Captures intrinsic functional brain connectivity without requiring an explicit task.
– Output: classify participants into:
  - ASD: Autism Spectrum Disorder
  - TD: Typically Developing

Models

- Baseline Models: + Support Vector Machine (SVM)
- Deep learning: + 2D CNN
- Brain Graph: Using GNN

Explainable AI

Model	XAI method	Explanation output
SVM	SHAP	Contribution of each FC feature or connection to the prediction
2D CNN	Integrated Gradients	Attribution map over the FC matrix
GCN	GNNExplainer	Important subgraph, nodes and functional connections

> XAI explains which brain-network features contribute to a model prediction; it does not establish causal mechanisms of ASD.

04

Conclusion

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Expected contributions:

- Systematic model comparison
  - Fair comparison of classical ML, deep learning and GNN models under a unified experimental design.
- Explainable brain-network analysis
  - XAI is used to identify important ROIs and functional connections and relate them to existing neurobiological literature.

Citation

[1] [4] Hodges, H., Fealko, C., & Soares, N. (2020). Autism spectrum disorder: definition, epidemiology, causes, and clinical evaluation.

[2] World Health Organization, ICD-11, 6A02 Autism spectrum disorder.

[3] American Psychiatric Association. (2013). Diagnostic and statistical manual of mental disorders (5th ed.). American Psychiatric Publishing.

[5] CDC ADDM Network, 2025.

[6] Frye, R. E., Vassall, S., Kaur, G., Lewis, C., Karim, M., & Rossignol, D. (2019). Emerging biomarkers in autism spectrum disorder: A systematic review. Annals of Translational Medicine, 7(23), 792.

[7] Yu, Y., Ozonoff, S., & Miller, M. (2023). Assessment of autism spectrum disorder. Assessment, 31(1), 24–41.

[8] International Neuroimaging Data-sharing Initiative. (2016). Autism Brain Imaging Data Exchange